On the Guidance of Flow Matching: review on synthetic data and towards cell data

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Abstract

Flow matching has shown state-of-the-art performance in various generative tasks in image generation and scientific applicaiton. In many applicatons, it is desirable to generate specific parts of the target distribution which can be dynmaically controlled after training. Guidance methods have been proposed to steer a frozen flow matching model towards a defined objective. In this paper, we evaluate the guidance methods for flow matching on synthetic data and discuss the limiting assumptions of the methods. This is important to understand for potential applications in single-cell data.

1 Introduction

Flow matching is a generative method (Lipman et al., 2023).

2 Method

2.1 Coupling

2.2 Guidance

2.2.1 Discussion on guidance methods

The MC method needs only minimal assumptions and minimal tuning in terms of hyperparameter but still produces solid results. However, the MC method does not scale well to higher dimensions because the variance explodes.

The learned (GM) model performs best in terms of W2 distance and energy distance, but several models need to be trained (first disadvantage) and one has to do so for each new energy function (second disadvantage).

The disadvantage of the covariance methods are the additional assumptions that are made, which can be violated in practice.

How does the bias of the guidance methods look like? Maybe something like mode collapse in the Cov methods

2.2.2 Synthetic data experiments

We evaluate the guidance methods on synthetic data and compare the results in terms of W2 distance, energy distance, divergence trace ratio and off-manifold rate. The 2D synthetic data allows to have interpretable results and visualize the generated samples compared to other complex real-world tasks like image generation or single-cell data.

Preprint.

3 Results

3.1 Compare guidance methods

Please read the instructions below carefully and follow them faithfully.

$$J(x) = -\log \sum_{i \in 0,2,5} \exp\left(-\frac{|x - c_i|_2^2}{2\sigma^2}\right)$$

$$J(x) = \frac{|x - c_0|_2^2}{2\sigma^2} + \frac{|x - c_2|_2^2}{2\sigma^2} + \frac{|x - c_5|_2^2}{2\sigma^2}$$

Table 1: Comparison of methods across various metrics. The best method for each metric is **bolded**, and the second best is underlined.

Method	W2 ↓	Energy Dist ↓	Div Trace Ratio	Off-Manifold Rate ↓
Learned (GM)	**0.0802**	**0.0006**	0.9757	0.0659
Class-cond. CFM	0.1632	0.0033	1.0069	0.0786
MC	0.1698	0.0030	**0.9969**	0.0801
cov-G	0.2772	0.0296	1.2113	0.0068
sim-MC	0.3908	0.0739	0.7570	0.5098
cov-A	0.7905	0.2578	0.5410	**0.0000**
CFM	0.8852	0.3804	1.5355	0.6782

3.2 Strong Coupling: when the assumptions break

3.3 Tuning hyperparameters

We want to discuss the effects of the hyperparameters on the guidance methods.

3.3.1 ODE solver

We increased the number of steps in the ODE solver from 100 to 200. This increased the performance of the guidance methods. Moreover, one could use a different ODE solver, e.g. Runge-Kutta 4th order, which might improve the generated samples further with the downside of increased computational cost. An early stopping criterion could be used to mitingate the problem of numerical instabilities at $t = 1$ and mode collapse in the MC guidance method.

3.3.2 MC Guidance

The parameter `mc_ep` (ϵ) appears in the denominator $(1 - t + \epsilon)$ of both the MC importance weights and the velocity correction $u = (x_1 - x_t)/(1 - t + \epsilon)$, acting as a numerical stabilizer as $t \to 1$. Without sufficient ϵ, back-projected source points fall outside the prior support, causing all importance weights to clamp to a uniform distribution and injecting large random forces that scatter particles off the manifold. We tried $\epsilon \in \{0.001, 0.005, 0.01, 0.05, 0.1\}$, with results shown in Table 2. Hence, we increased `mc_ep` from 0.001 to 0.05, which reduced the numerical instability and the outliers of the generated distribution significantly.

Table 2: Effect of the `mc_ep` parameter ϵ on sample quality.

ϵ	W_2 Distance ↓	Off-Manifold Rate ↓
0.001 (default)	0.4244	0.266
0.010	0.3053	0.042
0.050 (tuned)	**0.1091**	**0.002**
0.100	0.2032	0.004

3.3.3 Cov Guidance

4 Conclusion

Conclusion

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References

Ruiqi Feng, Chenglei Yu, Wenhao Deng, Peiyan Hu, and Tailin Wu. On the guidance of flow matching. In Aarti Singh, Maryam Fazel, Daniel Hsu, Simon Lacoste-Julien, Felix Berkenkamp, Tegan Maharaj, Kiri Wagstaff, and Jerry Zhu, editors, *Proceedings of the 42nd International Conference on Machine Learning*, volume 267 of *Proceedings of Machine Learning Research*, pages 16993–17029. PMLR, 13–19 Jul 2025. URL https://proceedings.mlr.press/v267/feng25s.html.

Dominik Klein, Théo Uscidda, Fabian Theis, and Marco Cuturi. Genot: Entropic (gromov) wasserstein flow matching with applications to single-cell genomics. In A. Globerson, L. Mackey, D. Belgrave, A. Fan, U. Paquet, J. Tomczak, and C. Zhang, editors, *Advances in Neural Information Processing Systems*, volume 37, pages 103897–103944. Curran Associates, Inc., 2024. doi: 10.52202/079017-3301. URL https://proceedings.neurips.cc/paper_files/paper/2024/file/bc46e29f91e676747c584ca181cb0ea1-Paper-Conference.pdf.

Yaron Lipman, Ricky T. Q. Chen, Heli Ben-Hamu, Maximilian Nickel, and Matthew Le. Flow matching for generative modeling. In *The Eleventh International Conference on Learning Representations*, 2023. URL https://openreview.net/forum?id=PqvMRDCJT9t.

Alexander Tong, Kilian FATRAS, Nikolay Malkin, Guillaume Huguet, Yanlei Zhang, Jarrid Rector-Brooks, Guy Wolf, and Yoshua Bengio. Improving and generalizing flow-based generative models with minibatch optimal transport. *Transactions on Machine Learning Research*, 2024. ISSN 2835-8856. URL https://openreview.net/forum?id=CD9Snc73AW. Expert Certification.

A Appendix

Appendix